Ali Hamza Malik

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Skills

Programming Languages: Python, C/C++, Julia, Rust, JavaScript, Kotlin, MATLAB
Formal Methods: Model Checking (TLA+, nuXMV), Theorem Proving (RCoq), Protocol Verification (Tamarin, CryptoVerif), SMT Solving & Encoding (Z3, Bitwuzla), Symbolic Execution, State Machine Modeling
Generative AI/ML: Huggingface, Transformers, PyTorch, TensorFlow, Keras, SFT, DPO, RLHF, RLAF (RL-Automated Feedback), LoRA, QLoRA, RAG, Specification-Aware Fine-Tuning, Agentic AI
DevOps & Cloud: Git/GitHub, Docker, Shell Scripting, CMake, AWS (EC2, Route 53), MongoDB, N8N

Relevant Experience

Graduate Research Assistant, Khwarizmi Lab September 2023 – Present
University of Massachusetts Amherst

- Discovered previously undocumented security flaws in quantum key distribution (QKD) protocols under quantum-classical interactions by developing a Tamarin prover-based verification framework
- Uncovered 3 critical access control vulnerabilities in U.S. ACH direct payment systems by formally modeling banking payment protocols in TLA+ to perform bounded model checking

Undergraduate Research Assistant, Communication Systems and Networks Lab September 2022 – July 2023
National University of Sciences & Technology

- Built an event-driven coordination system using C++ and MAVLink for multi-UAV aerial swarms on Raspberry Pi and Pixhawk hardware stack to enable autonomous multi-agent mission planning
- Optimized leader–follower formation control algorithms (flock, line, helical) with dynamic reconfiguration, reducing formation-switching latency to under 2 minutes

Hardware Security Intern, IC Design Lab June 2022 – September 2022
National University of Sciences & Technology

- Developed ENIGMA, an automated Python framework for inserting logic-locking defenses into Verilog HDL designs for reducing IP reverse-engineering risk
- Developed SAT-based attack algorithms that successfully recovered logic-locking keys from obfuscated circuits in under 6 hours to expose weaknesses in 2 commercial logic-locking schemes

Machine Learning Intern, TUKL Deep Learning Lab June 2021 - September 2021
National University of Sciences & Technology

- Built an automated ETL pipeline using Python and regex to process 10,000+ PDF raw court documents to fine-tune NLP transformer models (BERT and RoBERTa) for verdict prediction of court cases

Education

PhD in Electrical & Computer Engineering Expected May 2028
University of Massachusetts Amherst

Teaching Assistant: Junior Design Project, Fundamentals of Electrical Engineering, Intro to Programming

BE in Electrical Engineering *and* Minor in Computer Science Completed June 2023
National University of Sciences and Technology

Capstone Project: Optimized security-cost tradeoffs of hardware obfuscation on a proprietary RISC-V microprocessor
Funded Project: Grant-funded research implementing a low-latency control system for multi-UAV aerial swarms

Relevant Projects

LogicTune — *Python, PyTorch, LLM, RLAF, DPO, LoRA, Model Checking*

- Improved autonomous driving safety from 60% to 90% with DPO and feedback from model checking

Custom RV32I Processor Core — *Verilog HDL (RTL), FPGA (DE1-SoC), GCC RV32I cross-compiler, CoreMark benchmarks*

- Designed and implemented a complete 32-bit RISC-V processor in Verilog RTL from first principles

Honors and Awards

2023 - Rector's Gold Medal: Awarded for best senior project, National University of Sciences & Technology
2022 - 2nd Place, CSAW'22 LLC: For a global security hackathon competition, NYU School of Engineering